Quiz | The limit function of th | The uniform converge | A sequence of integral

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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 10:06 AM, 21 November 2023

Question # 10 of 10 (Start time: 10:15:20 AM, 21 November 2023)

Total Marks: 1

A sequence of integrable functions converging uniformly to an integrable function, the limit of integrals is -------equal to the integral of limit function.

If a sequence of integrable functions converges uniformly to an integrable function, then the limit of the integrals is necessarily equal to the integral of the limit function. The uniform convergence property allows the interchange of the limit and integral operations under certain conditions, ensuring that the limit of the integrals is indeed the integral of the limit function.

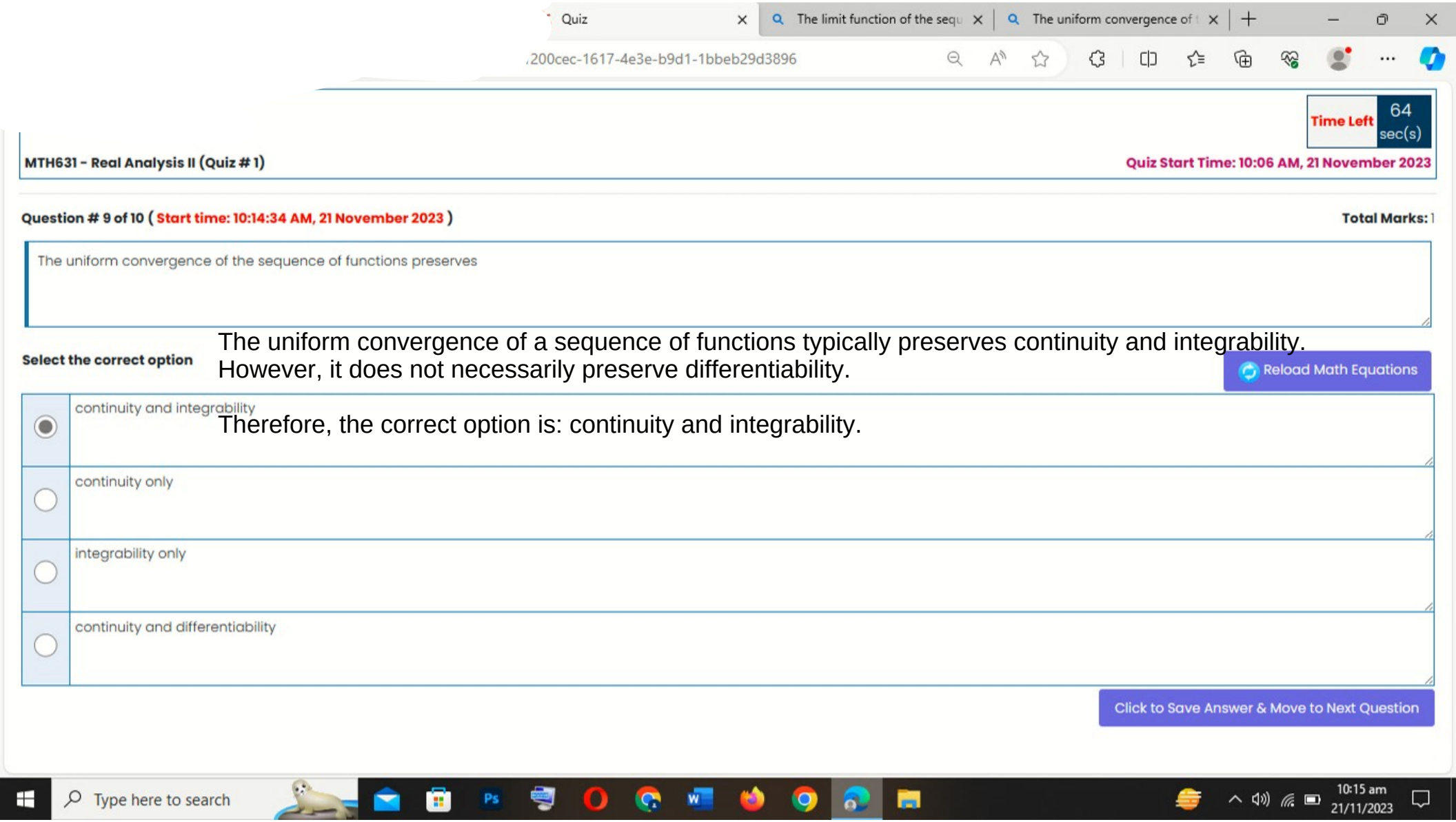
Select the correct option

- (●) necessarily
- ( ) not

Click to Save Answer & Move to Next Question

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10:16 am
21/11/2023


Quiz
The limit function of the sequ
The uniform convergence of t
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MTH631 - Real Analysis II (Quiz # 1)
Quiz Start Time: 10:06 AM, 21 November 2023
Question # 9 of 10 (Start time: 10:14:34 AM, 21 November 2023)
Total Marks: 1
The uniform convergence of the sequence of functions preserves
Select the correct option
The uniform convergence of a sequence of functions typically preserves continuity and integrability. However, it does not necessarily preserve differentiability.
Reload Math Equations
continuity and integrability
Therefore, the correct option is: continuity and integrability.
continuity only
integrability only
continuity and differentiability
Click to Save Answer & Move to Next Question
Type here to search
10:15 am
21/11/2023

Quiz | The limit function of the sequ | The method of testing the un
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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 10:06 AM, 21 November 2023

Question # 8 of 10 (Start time: 10:13:45 AM, 21 November 2023) **Total Marks:** 1

The method of testing the uniform convergence of a series $\sum f_j$ of functions by definition is hard because - - - .

The correct option is: finding the sn (nth partial sum) is not always easy.

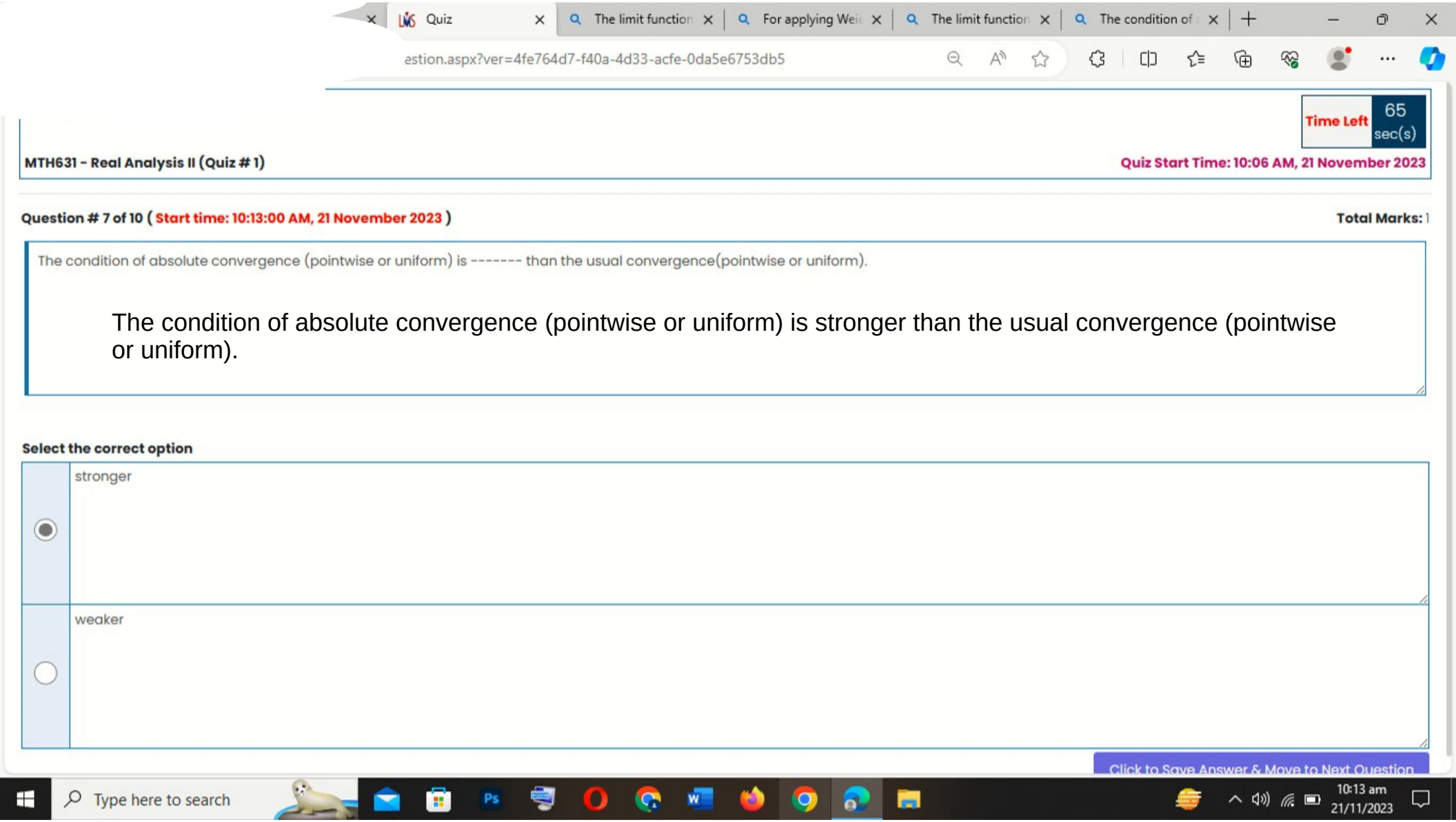
Select the correct option

Reload Math Equations

- (●) finding the S_n (nth partial sum) is not always easy
- ( ) checking continuity of each f_j is not an easy task

Click to Save Answer & Move to Next Question

Type here to search 10:14 am 21/11/2023


Quiz
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For applying Wei
The limit function
The condition of
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MTH631 - Real Analysis II (Quiz # 1)
Quiz Start Time: 10:06 AM, 21 November 2023
Question # 7 of 10 (Start time: 10:13:00 AM, 21 November 2023)
Total Marks: 1
The condition of absolute convergence (pointwise or uniform) is ------- than the usual convergence(pointwise or uniform).
The condition of absolute convergence (pointwise or uniform) is stronger than the usual convergence (pointwise or uniform).
Select the correct option
stronger
weaker
Click to Save Answer & Move to Next Question
Type here to search
10:13 am
21/11/2023

The limit function of th | For applying Weierstra | The limit function of th
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Time Left 51 sec(s)

MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 10:06 AM, 21 November 2023

Question # 6 of 10 (Start time: 10:11:49 AM, 21 November 2023) **Total Marks:** 1

The limit function of the sequence $\{F_n(x)\}$ defined by;
$F_n(x) = 1 + x + x^2 + \cdots + x^{n-1}, x \in [0,1)$ is - - - - .

Select the correct option

Reload Math Equations

- (●) not defined
- ( ) bounded
- ( ) unbounded
- ( ) discontinuous in the given domain

the correct option is: not defined.

Click to Save Answer & Move to Next Question

Type here to search
10:12 am
21/11/2023

Quiz | The limit function of the sequ | For applying Weierstrass' M -

...ion.aspx?ver=7093c190-2f14-463c-b256-c46c8d1a8a5d

Time Left 55 sec(s)

MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 10:06 AM, 21 November 2023

Question # 5 of 10 (Start time: 10:10:54 AM, 21 November 2023)

Total Marks: 1

For applying Weierstrass' M - test, which of the following can be taken as dominating series for the uniform convergence of $\sum_{n=1}^{\infty} \frac{e^{-2nx}}{4n^2-1}, x \geqslant 0$?

the correct option is:
summation 1/4n^2-1

Select the correct option

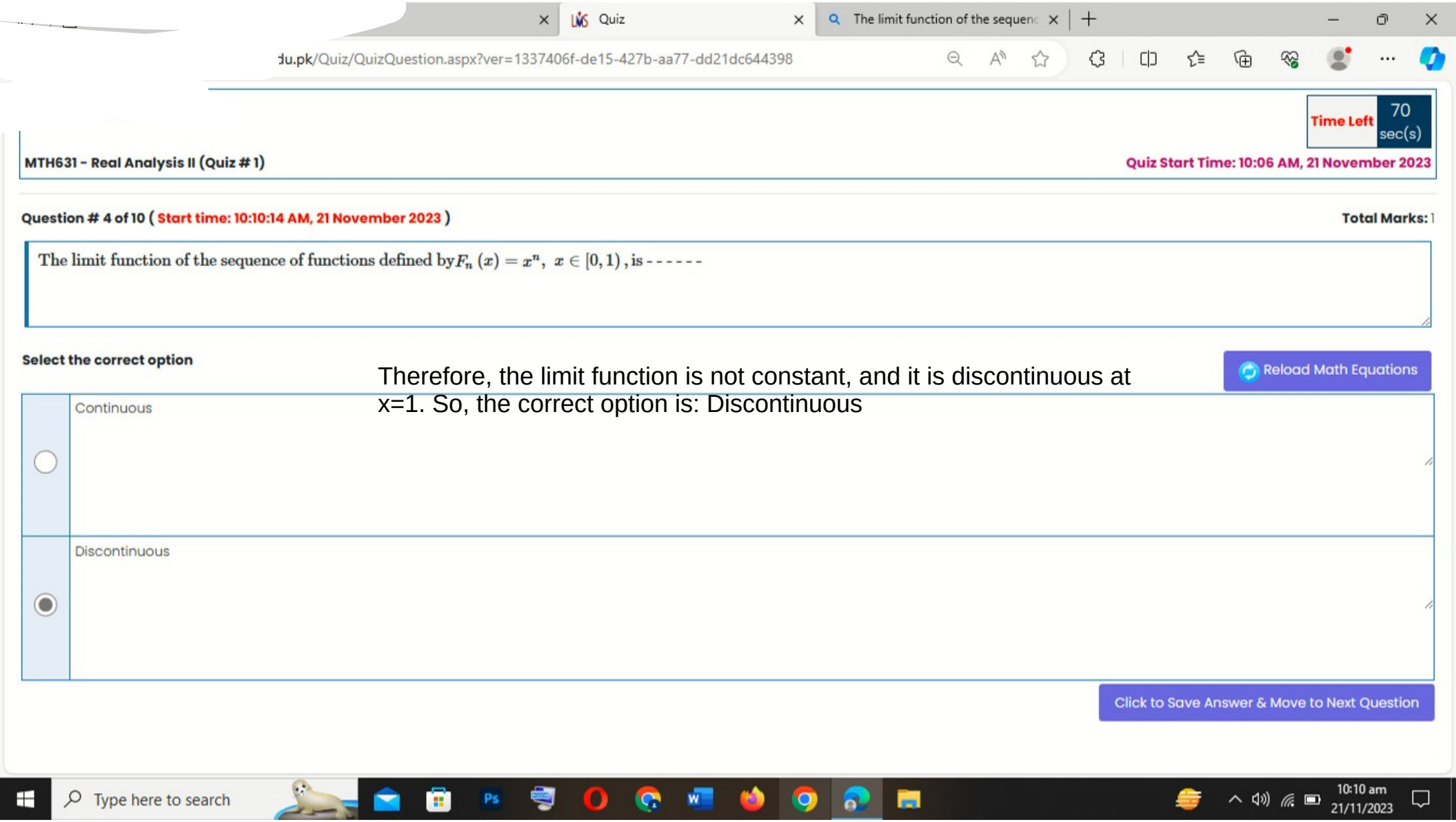
Reload Math Equations

- (●) $\sum \frac{1}{4n^2-1}$
- ( ) $\sum \frac{1}{4n^2}$
- ( ) $\sum \frac{1}{e^{2nx}}$
- ( ) $\sum \frac{1}{e^{-2nx}}$

Click to Save Answer & Move to Next Question

Type here to search

10:11 am
21/11/2023



Therefore, the limit function is not constant, and it is discontinuous at x=1. So, the correct option is: Discontinuous

Time Left 16 sec(s)

MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 10:06 AM, 21 November 2023

Question # 3 of 10 (Start time: 10:08:24 AM, 21 November 2023) **Total Marks:** 1

The power series expansion of $f(x) = \sin x$ about $x_0 = 0$ is

Select the correct option

Reload Math Equations

- $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty.$
- $\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty,$
- $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty,$ ✓
- $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty,.$

Click to Save Answer & Move to Next Question

Quiz | For the sequence of functions | If the sequence of functions

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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 10:06 AM, 21 November 2023

Question # 2 of 10 (Start time: 10:07:33 AM, 21 November 2023)

Total Marks: 1

If the sequence of functions defined and converges pointwise as,

$$\lim_{n\to\infty} F_n(x) = \lim_{n\to\infty}\left(1-\frac{x_n}{n}\right) = 1 = F(x), 0 \leqslant x \leqslant 1.$$

Then the value of $\sup_{x\in[0,1]} |F_n(x) - F(x)| = 0 \Rightarrow \{F_n(x)\}$ converges - - - - .

Select the correct option

Reload Math Equations

- ( ) not uniformly
- (•) uniformly

the correct option is: uniformly.

Click to Save Answer & Move to Next Question

Type here to search

10:08 am
21/11/2023

Quiz | For the sequence of functions {Fr

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Time Left 55 sec(s)

MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 10:06 AM, 21 November 2023

Question # 1 of 10 (Start time: 10:06:32 AM, 21 November 2023) **Total Marks:** 1

For the sequence of functions $\{F_n(x)\}$ defined and converges pointwise as;

$F_n(x) = \frac{nx}{1+n^2x^2}, x \in [0,1]$, the value of $\lim_{n\to\infty} \|F_n(x) - F(x)\|_{[0,1]} \Rightarrow \{F_n(x)\}$ is $----$, where $\lim_{n\to\infty} F_n(x) = F(x)$.

Select the correct option

Reload Math Equations

- (●) uniformly convergent
- ( ) not uniformly convergent

correct option is: uniformly convergent.

Click to Save Answer & Move to Next Question

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10:07 am
21/11/2023

lms login vu - Sea | Quiz | To check uniform | The limit function | Let f be defined b
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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 10 of 10 (Start time: 09:49:37 AM, 21 November 2023) **Total Marks:** 1

Let f be defined by a power series with finite radius of convergence R. Then $\lim_{x\to(x_0+R)^-} f(x) = \sum_{n=0}^{\infty} a_n R^n$ only if

Select the correct option

Reload Math Equations

- (●) $\sum_{n=0}^{\infty} a_n R^n$ converges
- ( ) None of these.
- ( ) $\sum_{n=0}^{\infty}(-1)^n a_n R^n$ converges.
- ( ) $\sum_{n=0}^{\infty} b_n R^n$ is uniformly convergent.

Click to Save Answer & Move to Next Question

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Quiz | To check | The limit | For |x|≥1 | If f(x)=∑ | The unif | A power
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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 9 of 10 (Start time: 09:49:00 AM, 21 November 2023) **Total Marks:** 1

A power series is continuous and differentiable in

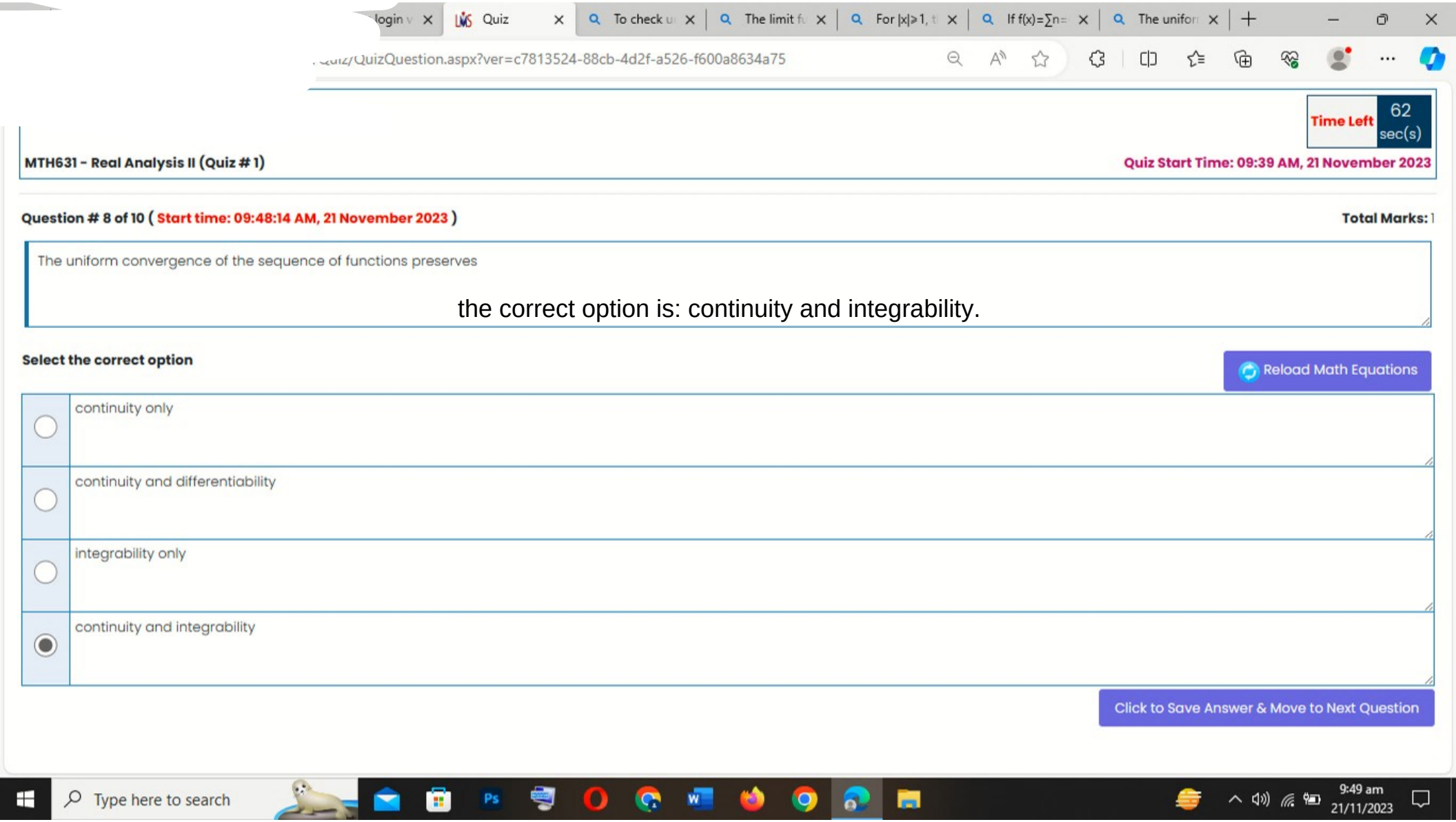
Select the correct option

Reload Math Equations

- (●) its interval of convergence.
- ( ) in every interval $[x_0 - r, x_0 + r]$ such that $0 < r$.

Click to Save Answer & Move to Next Question

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9:49 am
21/11/2023


Quiz
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If f(x)=∑n=
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MTH631 - Real Analysis II (Quiz # 1)
Quiz Start Time: 09:39 AM, 21 November 2023
Question # 8 of 10 (Start time: 09:48:14 AM, 21 November 2023)
Total Marks: 1
The uniform convergence of the sequence of functions preserves
the correct option is: continuity and integrability.
Select the correct option
Reload Math Equations
continuity only
continuity and differentiability
integrability only
continuity and integrability
Click to Save Answer & Move to Next Question
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9:49 am
21/11/2023

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...(Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 7 of 10 (Start time: 09:46:18 AM, 21 November 2023) **Total Marks:** 1

If $f(x)=\sum_{n=0}^{\infty} a_n(x-x_0)^n$, having $R>0$ as a radius of convergence, and $f(x)$ is continuous and differentiable in its intervel of convergence, then radius of convergence of $f'(x)$ is - - - - - - .

Select the correct option

the correct option is: also R

Reload Math Equations

○	$R+h, h>0$
◉	also R
○	$\frac{1}{R}, R\neq 0$
○	$R-h, h<0$

Click to Save Answer & Move to Next Question

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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 6 of 10 (Start time: 09:44:52 AM, 21 November 2023) **Total Marks:** 1

For $|x| \geqslant 1$, the series: $1 + x + x^2 + \cdots + x^{n-1}$ $- - - - - - - - as\ n \to \infty.$

the correct option is: converges for
|x|<x and diverges for |x|>=1

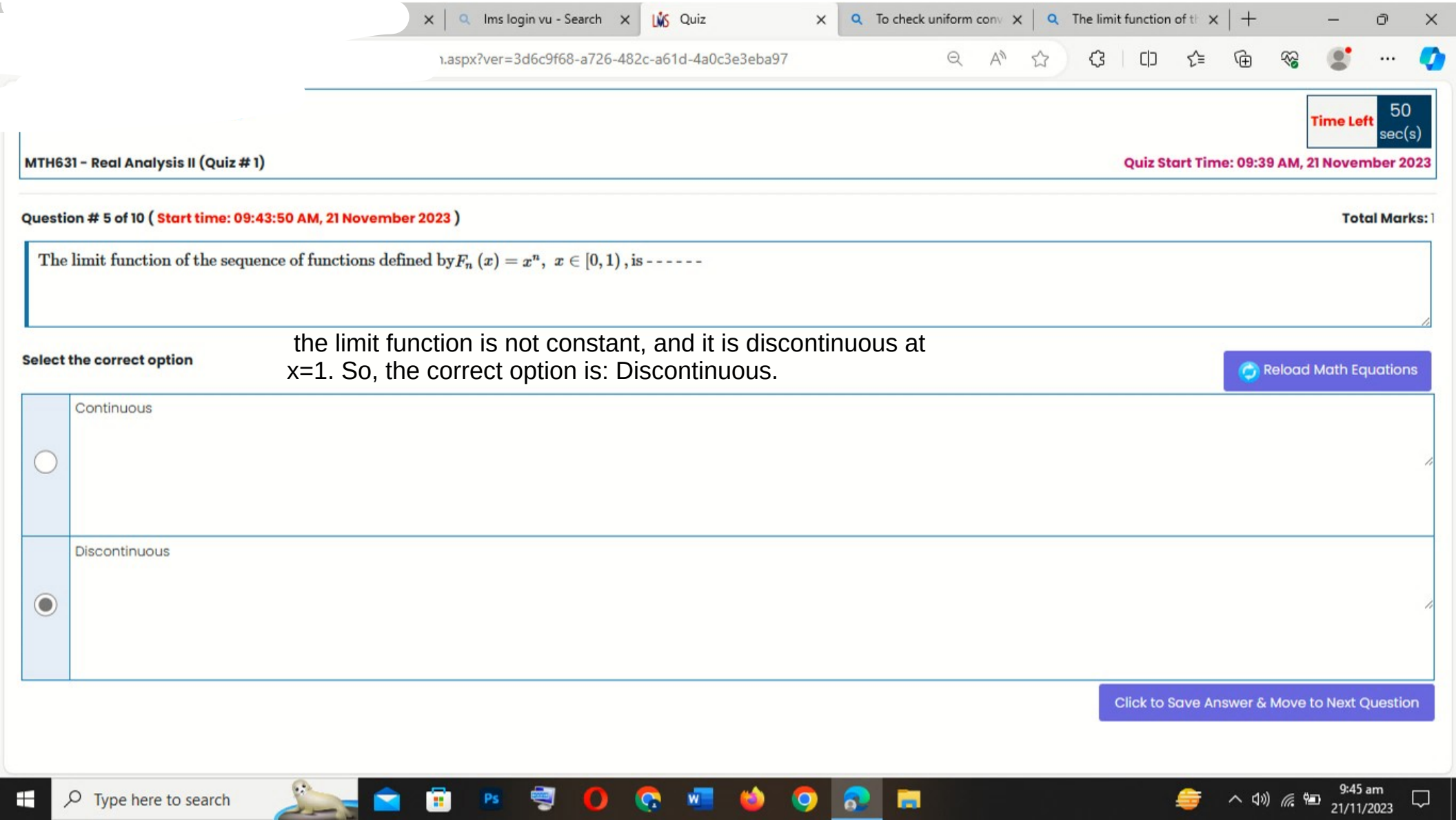
Select the correct option

Reload Math Equations

- (●) diverges
- ( ) converges
- ( ) converges uniformly
- ( ) converges pointwise

Click to Save Answer & Move to Next Question

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9:46 am
21/11/2023


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MTH631 - Real Analysis II (Quiz # 1)
Quiz Start Time: 09:39 AM, 21 November 2023
Question # 5 of 10 (Start time: 09:43:50 AM, 21 November 2023)
Total Marks: 1
The limit function of the sequence of functions defined by $F_n(x) = x^n,\ x \in [0,1)$, is - - - - - -
Select the correct option
the limit function is not constant, and it is discontinuous at x=1. So, the correct option is: Discontinuous.
Reload Math Equations
Continuous
Discontinuous
Click to Save Answer & Move to Next Question
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9:45 am
21/11/2023

ogin vu | Quiz | To check unif | The limit func | For the seque | Let f be defin
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Time Left 34 sec(s)

MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 4 of 10 (Start time: 09:42:23 AM, 21 November 2023) **Total Marks:** 1

Let f be defined by a power series with finite radius of convergence R. Then $\lim_{x \to (x_0+R)+} f(x) = \sum_{n=0}^{\infty} a_n R^n$ only if

Select the correct option

Reload Math Equations

- None of the above
- $\sum_{n=0}^{\infty} a_n R^n$ converges. True
- $\sum_{n=0}^{\infty} (-1)^n a_n R^n$ converges
- $\sum_{n=0}^{\infty} b_n R^n$ is convergent.

Click to Save Answer & Move to Next Question

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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 3 of 10 (Start time: 09:41:43 AM, 21 November 2023)

Total Marks: 1

For the sequence of functions $\{F_n(x)\}$ defined and converges pointwise as;

$$\lim_{n\to\infty} F_n(x) = \frac{\sin nx}{\sqrt{n}} = 0 = F(x), x \in [0, \pi],$$

the value of $\lim_{n\to\infty} \|F_n(x) - F(x)\|_{[0,1]} \Rightarrow \{F_n(x)\}$ is $----$.

Select the correct option

the correct option is: uniformly convergent.

Reload Math Equations

- (●) uniformly convergent
- ( ) non-uniformly convergent

Click to Save Answer & Move to Next Question

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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 2 of 10 (Start time: 09:40:34 AM, 21 November 2023) **Total Marks:** 1

The limit function of the sequence $\{F_n(x)\}$ defined by;
$F_n(x) = 1 + x + x^2 + \cdots + x^{n-1}, x \in [0,1)$ is - - - - .

Select the correct option

Reload Math Equations

- (●) discontinuous in the given domain
- ( ) unbounded
- ( ) not defined
- ( ) bounded

Click to Save Answer & Move to Next Question

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MTH631 - Real Analysis II (Quiz # 1)

Quiz Start Time: 09:39 AM, 21 November 2023

Question # 1 of 10 (Start time: 09:39:27 AM, 21 November 2023) **Total Marks:** 1

To check uniform convergence of $\sum \frac{\sin nx}{n^2}$ we can use and apply _____. Choose the best answer.

The correct option is: Weierstrass M-test.

Select the correct option

Reload Math Equations

- ○ dominated convergence test
- ○ non of these
- ○ both a and b
- ◉ Weierstrass M-test

Click to Save Answer & Move to Next Question

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